Module 9: Outlier Detection

- Basic concepts
- Statistical approaches
- Proximity-based approaches
- Reconstruction-based approaches
- Clustering and classification-based approaches
- Mining contextual and collective outliers
- Outlier detection in high-dimensional data

Motivation

- How to detect suspicious credit card transactions, such as transactions of unusual amounts and multiple onsite transactions within 10 minutes in two locations hundreds of miles away?
- Most transactions are normal and only very few are anormal
- Outlier detection (also known as anomaly detection) is the process of finding outliers or anomalies, the data objects with behaviors that are very different from expectation
- Many applications, such as medical care, public safety and security, industry damage detection, image processing, sensor and video network surveillance, national security, and system intrusion detection

Outlier Detection versus Clustering

- Two highly related but different tasks
- Difference in purpose
  - Clustering finds the majority patterns in a data set and organizes the data accordingly
  - Outlier detection tries to capture those exceptional cases that deviate substantially from the majority patterns
- Difference in methodology
  - Outlier detection might also use supervision during the detection process
  - Clustering analysis is typically unsupervised in nature